Finite-Volume Derivation of TRISO Diffusion and Release

Supervisor Delivery Package

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Symbols and units

Symbol	Meaning	Units
$r,\ t$	radius, time	m, s
R_i	outer radius of region i (Kernel, Buffer, IPyC, SiC, OPyC), $R_0 = 0$, $R_5 = R$	m
D_i	diffusivity of region i (constant for a fixed nuclide and temperature)	$\mathrm{m^2\,s^{-1}}$
$c(r,t)$	concentration of the tracked species	$\mathrm{mol\,m^{-3}}$
J_r	outward radial flux	$\mathrm{mol\,m^{-2}\,s^{-1}}$
S	volumetric source	$\mathrm{mol\,m^{-3}\,s^{-1}}$
c_0	initial kernel concentration	$\mathrm{mol\,m^{-3}}$
$h,\ c_\infty$	Robin transfer coefficient, external concentration (Case B only)	$\mathrm{m\,s^{-1}}$, $\mathrm{mol\,m^{-3}}$
$\Lambda_n,\ \phi_n$	n-th eigenvalue and eigenfunction	$\mathrm{s^{-1}}$, $\mathrm{mol\,m^{-3}}$
$k_i = \sqrt{\Lambda/D_i}$	wave number in region i	$\mathrm{m^{-1}}$
$u = r\phi,\ q = Du'$	transformed eigenfunction and its flux variable	$\mathrm{mol\,m^{-2}}$, $\mathrm{mol\,m^{-1}\,s^{-1}}$
$V_j,\ A_f$	cell volume, face area	$\mathrm{m^3}$, $\mathrm{m^2}$
G_f	face conductance	$\mathrm{m^3\,s^{-1}}$
$M,\ M_{\mathrm{rel}},\ F$	inventory in the particle, cumulative release, release fraction M_{rel}/M^0	mol, mol, 1

1 Scope and assumptions

1. Five concentric regions: Kernel, Buffer, IPyC, SiC and OPyC. Region i occupies $R_{i-1} < r < R_i$.
2. Spherical symmetry. In each region D_i is constant.
3. Interfaces are ideal: concentration and normal diffusive flux are continuous (no partition coefficient).
4. **Case A (release).** Frozen initial kernel inventory, no source for $t > 0$, absorbing outer surface. The finite-volume (FV) solver and the exact series of Section 4 both solve Case A with identical inputs, so they are compared directly.

5. **Case B (analytical benchmark).** Continuous generation with a Robin outer boundary. Only its homogeneous eigenvalue condition is implemented (Section 4.6). Case A and Case B must not be compared with each other.

2 From conservation to the spherical equation

For any fixed control volume V with outward unit normal $\mathbf{n}$,

$$\frac{d}{dt}\int_V c\,dV = -\int_{\partial V}\mathbf{J}\cdot\mathbf{n}\,dA + \int_V S\,dV. \tag{TRISO-GOV-020}$$

The divergence theorem turns the surface term into a volume integral,

$$\int_V \left(\frac{\partial c}{\partial t} + \nabla\cdot\mathbf{J} - S\right) dV = 0,$$

and because V is arbitrary the integrand vanishes:

$$\frac{\partial c}{\partial t} + \nabla\cdot\mathbf{J} = S. \tag{TRISO-GOV-030}$$

Fick's law

$$\mathbf{J} = -D\nabla c \tag{TRISO-GOV-034}$$

gives

$$\frac{\partial c}{\partial t} = \nabla\cdot(D\nabla c) + S. \tag{TRISO-GOV-040}$$

With spherical symmetry $\mathbf{J} = J_r(r,t)\,\mathbf{e}_r$. A balance on the shell $r < r' < r + dr$ gives $\partial_t(4\pi r^2 c\,dr) = -\left[4\pi(r+dr)^2 J_r(r+dr) - 4\pi r^2 J_r(r)\right] + 4\pi r^2 S\,dr$, so $\nabla\cdot\mathbf{J} = r^{-2}\partial_r(r^2 J_r)$ and, in region i,

$$\boxed{\frac{\partial c_i}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_i \frac{\partial c_i}{\partial r}\right) + S_i} \tag{TRISO-FV-100}$$

3 Conditions

Centre (regularity and symmetry):

$$\left.\frac{\partial c_1}{\partial r}\right|_{r=0} = 0.$$

Interfaces $r = R_i$, $i = 1,\ldots,4$ (two conditions each):

$$c_i(R_i,t) = c_{i+1}(R_i,t), \qquad D_i\left.\frac{\partial c_i}{\partial r}\right|_{R_i^-} = D_{i+1}\left.\frac{\partial c_{i+1}}{\partial r}\right|_{R_i^+}.$$

Outer surface, Case A (absorbing) and Case B (Robin):

$$c_5(R,t) = 0 \qquad \text{or} \qquad -D_5\left.\frac{\partial c_5}{\partial r}\right|_R = h\,[c_5(R,t) - c_\infty].$$

Initial data and source, Case A:

$$c(r,0) = \begin{cases} c_0, & 0 \le r < R_1 \\ 0, & R_1 < r < R \end{cases} \qquad S_i = 0 \ (t > 0).$$

4 Exact solution of Case A

4.1 Separation and transformation

Set $c = \phi(r)\, e^{-\Lambda t}$. Region by region (the product $r^2 D\, \phi'$ is the conserved flux quantity),

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 D_i \frac{d\phi}{dr}\right) + \Lambda\phi = 0.$$

With $u = r\phi$ (Equation TRISO-ML-431) this becomes, in each region,

$$u'' + k_i^2 u = 0, \qquad k_i^2 = \frac{\Lambda}{D_i}, \tag{TRISO-ML-432}$$

$$u_i(r) = A_i \sin(k_i r) + B_i \cos(k_i r), \qquad \phi_i = \frac{u_i}{r}. \tag{TRISO-ML-433}$$

Regularity at the centre removes $B_1 \cos(k_1 r)/r$, so $B_1 = 0$ (TRISO-ML-435).

4.2 Conditions in the transformed variable

Concentration continuity $\phi_i(R_i) = \phi_{i+1}(R_i)$ gives, after multiplying by R_i,

$$u_i(R_i) = u_{i+1}(R_i). \tag{TRISO-ML-439}$$

Flux continuity uses $\phi' = (u' - u/r)/r$:

$$D_i\left[u_i'(R_i) - \frac{u_i(R_i)}{R_i}\right] = D_{i+1}\left[u_{i+1}'(R_i) - \frac{u_{i+1}(R_i)}{R_i}\right]. \tag{TRISO-ML-446}$$

The outer condition for Case A is $u_5(R) = 0$.

Caution. The quantity $D\,u'$ is not continuous across an interface when the diffusivities differ. Matching $D\,u'$ instead of Equation TRISO-ML-446 gives wrong eigenvalues and wrong release curves.

4.3 Characteristic function

Carry the state (u, q) with $q = D\,u'$ from the centre outward. Inside a region of thickness L (solution of Equation TRISO-ML-432),

$$\begin{pmatrix} u \\ q \end{pmatrix}_{\text{out}} = \begin{pmatrix} \cos kL & \dfrac{\sin kL}{Dk} \\ -Dk \sin kL & \cos kL \end{pmatrix} \begin{pmatrix} u \\ q \end{pmatrix}_{\text{in}}. \tag{TRISO-DEL-010}$$

Across the interface at R_i, Equation TRISO-ML-439 keeps u and Equation TRISO-ML-446 gives

$$q_{i+1}(R_i) = q_i(R_i) + (D_{i+1} - D_i)\,\frac{u(R_i)}{R_i}. \tag{TRISO-DEL-011}$$

Start from $(u, q) = (0, 1)$ at $r = 0$ (regular solution), propagate through all five regions, and define

$$g(\Lambda) = u(R; \Lambda), \qquad g(\Lambda_n) = 0. \tag{TRISO-DEL-012}$$

The roots of g are the eigenvalues. This is the same condition as $\det \mathbf{M}(\Lambda) = 0$ (TRISO-ML-474) for the global coefficient system, evaluated by propagation instead of by a determinant.

Completeness of the root list. The number of eigenvalues not exceeding Λ equals the number of zeros of the regular solution in $(0, R]$ (Sturm oscillation count). Isolating roots by bisection on this integer count cannot skip a root.

4.4 Orthogonality and release series

With weight r^2 for ϕ, that is weight 1 for u, the interface and boundary terms cancel and (TRISO-ML-528, TRISO-ML-531)

$$\int_0^R u_m u_n \, dr = 0 \qquad (m \neq n).$$

Expanding the initial data, $c = \sum_n A_n \phi_n e^{-\Lambda_n t}$ with $A_n = c_0 \, I_n^{(1)}/N_n$, where

$$I_n^{(1)} = \int_0^{R_1} r \, u_n \, dr, \qquad I_n = \int_0^R r \, u_n \, dr, \qquad N_n = \int_0^R u_n^2 \, dr. \tag{TRISO-DEL-020}$$

The inventory is $M(t) = 4\pi \sum_n A_n I_n e^{-\Lambda_n t}$ and $M^0 = \frac{4\pi}{3} c_0 R_1^3$. Hence

$$\begin{aligned} F(t) &= 1 - \sum_n w_n \, e^{-\Lambda_n t}, \\ w_n &= \frac{3 \, I_n^{(1)} I_n}{R_1^3 \, N_n}, \qquad \sum_n w_n = 1. \end{aligned} \tag{TRISO-DEL-021}$$

The integrals are evaluated region by region from the propagated states, using $u(r) = u \cos ks + (q/Dk) \sin ks$ with $s = r - R_{i-1}$.

Homogeneous check. For equal D, $u_n = \sin(n\pi r/R)$, $\Lambda_n = n^2\pi^2 D/R^2$ and $N_n = R/2$. When the initial data fill the whole sphere ($R_1 = R$) this gives $w_n = 6/(n^2\pi^2)$, the classical result $F = 1 - \frac{6}{\pi^2} \sum n^{-2} e^{-n^2\pi^2 Dt/R^2}$.

4.5 Status and numerical limits

- **Completeness and truncation.** Orthogonality and the interface and boundary cancellations are derived in the foundation document. Completeness of the eigenfunctions for this singular piecewise problem is not proved there, so the series is checked numerically against an independent implementation and through the weight sum $\sum_n w_n \to 1$. Modes are kept up to $\Lambda_{\max}$ and the series is used only for $t \geq 40/\Lambda_{\max}$, where omitted modes are suppressed by e^{-40}.

- **Conditioning.** When the flux $D(u' - u/r)$ at an interface is much smaller than $D\,u'$ and $D\,u/r$ (strongly contrasting adjacent diffusivities), the propagation loses digits by cancellation. The code reports the smallest value of $|D(u' - u/r)|/(|Du'| + |Du/r|)$ over modes and interfaces, and an accuracy floor of about 2.2×10^{-16} divided by that ratio. Differences between FV and the series below that floor are not resolved.

4.6 Homogeneous Robin sphere (Case B reference)

For $\phi = \sin(zr/R)/r$ and $\mathrm{Bi} = hR/D$, the condition $-D\phi'(R) = h\phi(R)$ gives

$$z \cot z = 1 - \mathrm{Bi}.$$

5 Conservative spherical finite volumes

5.1 Geometry

Cell j occupies $r_{j-1/2} < r < r_{j+1/2}$ with interfaces aligned to cell faces.

$$V_j = \frac{4\pi}{3}\left(r_{j+1/2}^3 - r_{j-1/2}^3\right), \qquad A_{j\pm1/2} = 4\pi r_{j\pm1/2}^2. \tag{TRISO-FV-101}$$

The unknown is the exact cell average $C_j = V_j^{-1}\int c\,4\pi r^2 dr$ (TRISO-FV-118) represented at the volume centroid

$$r_j = \frac{3}{4}\frac{r_{j+1/2}^4 - r_{j-1/2}^4}{r_{j+1/2}^3 - r_{j-1/2}^3}. \tag{TRISO-FV-109A}$$

5.2 Exact cell balance

Multiply Equation TRISO-FV-100 by $4\pi r^2$ and integrate over the cell. The derivative term integrates exactly, $4\pi\left[r^2 D\,\partial_r c\right]_{r_{j-1/2}}^{r_{j+1/2}}$. With $J_r = -D\,\partial_r c$ (TRISO-FV-114) this equals $A_{j-1/2}J_{j-1/2} - A_{j+1/2}J_{j+1/2}$, so

$$V_j\frac{dC_j}{dt} = A_{j-1/2}J_{j-1/2} - A_{j+1/2}J_{j+1/2} + S_jV_j. \tag{TRISO-FV-123}$$

This is exact before any approximation of the face fluxes.

5.3 Face fluxes

Inside one material, with $\delta r_{PE} = r_E - r_P$ (TRISO-FV-124 to TRISO-FV-128),

$$J_e \approx D\,\frac{C_P - C_E}{\delta r_{PE}}.$$

At a face between materials, let $\delta_P = r_f - r_P$, $\delta_E = r_E - r_f$ and let C_I be the (continuous) interface concentration. Assume C varies linearly over each half cell. The same flux crosses both halves:

$$J_f = D_P\frac{C_P - C_I}{\delta_P} = D_E\frac{C_I - C_E}{\delta_E}.$$

Eliminating C_I (TRISO-FV-133 to TRISO-FV-138) gives $C_P - C_E = J_f\,(\delta_P/D_P + \delta_E/D_E)$, hence the conductance

$$\boxed{A_fJ_f = G_f\,(C_P - C_E), \qquad G_f = \frac{A_f}{\delta_P/D_P + \delta_E/D_E}} \tag{TRISO-FV-140}$$

For $D_P = D_E$ this is the ordinary two-point form, and for equal half distances it contains the harmonic mean $2D_PD_E/(D_P + D_E)$. The linear half-cell assumption with the face area A_f in both halves is the approximation made by the scheme. Comparison with the exact solution of Section 4 shows second-order convergence, including across material interfaces.

5.4 Semi-discrete equation and conservation

Substituting Equation TRISO-FV-140 into Equation TRISO-FV-123, with west and east conductances G_W and G_E,

$$\boxed{V_j\frac{dC_j}{dt} = G_W\,(C_{j-1} - C_j) + G_E\,(C_{j+1} - C_j) + S_jV_j} \tag{TRISO-FV-155}$$

The flux $G_f(C_E - C_P)$ leaves one cell and enters its neighbour with opposite sign, so all internal fluxes cancel exactly in $\sum_j V_j C_j$.

5.5 Centre and outer boundary

The first cell has $r_{1/2} = 0$, so its west face area and G_W vanish. No ghost cell is needed.

For the absorbing surface $c(R,t) = 0$ and the last centroid r_N,

$$\begin{aligned} G_R &= \frac{4\pi R^2 D_5}{R - r_N}, \qquad \dot{M}_{\mathrm{rel}} = G_R C_N, \\ V_N \frac{dC_N}{dt} &= G_W(C_{N-1} - C_N) - G_R C_N. \end{aligned} \tag{TRISO-DEL-030}$$

5.6 Backward Euler

For time step Δt,

$$\begin{aligned} \frac{V_j}{\Delta t}\left(C_j^{n+1} - C_j^n\right) = G_W &\left(C_{j-1}^{n+1} - C_j^{n+1}\right) \\ &+ G_E\left(C_{j+1}^{n+1} - C_j^{n+1}\right) + S_j^{n+1} V_j, \end{aligned}$$

$$\boxed{\begin{aligned} &- G_W C_{j-1}^{n+1} + \left(\frac{V_j}{\Delta t} + G_W + G_E\right) C_j^{n+1} - G_E C_{j+1}^{n+1} \\ &\quad = \frac{V_j}{\Delta t} C_j^n + S_j^{n+1} V_j \end{aligned}} \tag{TRISO-DEL-040}$$

The matrix is tridiagonal with positive diagonal and non-positive off-diagonal entries and is strictly diagonally dominant, hence an M-matrix. For non-negative old data and source the new concentrations are non-negative.

Subtraction-free elimination. Let $v_j = V_j/\Delta t$ and $g_j^W = G_W$ of cell j (so $g_{j+1}^W = g_j^E$). Define $e_1 = v_1$ and $e_j = v_j + g_j^W e_{j-1}/(e_{j-1} + g_j^W)$, pivots $d_j = e_j + g_j^E$, forward sweep $y_j = (\mathrm{rhs}_j + g_j^W y_{j-1})/d_j$, and back substitution $C_j = y_j + (g_j^E/d_j)\, C_{j+1}$ with $C_N = y_N$. Every operation adds, multiplies or divides non-negative numbers. The textbook pivot $b_j - a_j c'_{j-1}$ subtracts nearly equal numbers when the Fourier number $D\Delta t/\Delta r^2$ is large and degrades the inventory balance to about 10^{-9}. The form above keeps it below 10^{-12} (observed $\le 3 \times 10^{-13}$).

For verification only, Forward Euler has the coefficient-positivity bound

$$\boxed{\Delta t \le \min_j \frac{V_j}{G_W + G_E}} \tag{TRISO-FV-300}$$

Backward Euler is the production scheme and has no such restriction.

5.7 Inventory, release, balance

$$M^n = \sum_j V_j C_j^n, \qquad M_{\mathrm{rel}}^{n+1} = M_{\mathrm{rel}}^n + \Delta t\, G_R C_N^{n+1},$$

$$F^n = \frac{M_{\mathrm{rel}}^n}{M^0}, \qquad \dot{F}^n = \frac{G_R C_N^n}{M^0}.$$

Summing Equation TRISO-DEL-040 over j cancels all internal conductances and leaves

$$M^{n+1} + M_{\mathrm{rel}}^{n+1} = M^0$$

up to linear-solver and rounding error. The release uses the same boundary flux as the balance, so the residual $M^n + M_{\mathrm{rel}}^n - M^0$ is an independent diagnostic.

6 Limitations

- Results obtained with the synthetic test inputs supplied with the code are numerical checks, not physical validation.
- Temperature and fluence dependence of D, source schedules and a production Robin release mode are not implemented.
- The exact series is trustworthy only above the accuracy floor that the code reports for each run. The floor grows roughly in proportion to the diffusivity contrast between adjacent layers.
- Completeness of the eigenfunction set for the singular piecewise problem is checked numerically, not proved.